

# Slideshow Generator

Generate and browse an ordered slideshow from image, video, and document files in a project's `slides/` folder. The generator maintains a Markdown slide list, an editable browser view, and a standalone read-only view.

## Requirements

- Python 3. No third-party Python packages are needed for generation or serving.
- A modern web browser to view the slideshow. Actual playback depends on browser support for the selected media format and codec.
- Optional: `Pillow` for image dimensions and EXIF metadata.
- Optional: `ffmpeg` and `ffprobe` on `PATH` to check and re-encode MP4 video codecs.

## Project Layout

Run the script from a project folder containing `slides/` :

```
my-slideshow/  
├─ generate_slideshow.py  
├─ slides.md (generated by the script)  
├─ slideshow-gen.html (generated and used by the script)  
└─ slides/  
    ├─ photo.jpg  
    ├─ clip.mp4  
    └─ index.html (generated by the script)
```

Put media directly in `slides/`; files in nested folders are not scanned. The generator does not create this folder, so create it before the first run. It recognizes these filename extensions (case-insensitive):

**Images:** `.png`, `.apng`, `.jpg`, `.jpeg`, `.jfif`, `.gif`, `.webp`, `.avif`, `.bmp`, `.ico`, `.svg`

**Video:** `.mp4`

**Documents:** `.pdf`, `.md` (`slides.md` is excluded)

PDFs and Markdown files are shown in an embedded browser document view. Extension recognition does not guarantee that every browser can display or play a file. For widest MP4 browser compatibility, use H.264 video and AAC audio.

## Generate and View

From the project folder, run:

```
python generate_slideshow.py
```

The script updates or creates `slides.md`, writes the editable `slideshow-gen.html` in the project folder and writes the read-only `slides/index.html` beside the media. By default it starts a local server at `http://127.0.0.1:8000/slideshow-gen.html` and opens the editable view in your browser. Stop the server with Ctrl+C.

Options:

```
python generate_slideshow.py --help  
python generate_slideshow.py --no-serve  
python generate_slideshow.py --serve-only  
python generate_slideshow.py --dir path/to/project --port 8080  
python generate_slideshow.py --fix-videos
```

- `--dir DIR` : use a project folder other than the current directory. It must contain `slides/`.
- `--port PORT` : choose the local HTTP server port (default `8000`).
- `--no-serve` : generate/update the list and both HTML files, then exit.

- `--serve-only` : serve existing `slides.md` and `slideshow-gen.html` without regenerating them. The project must also contain `slides/`.
- `--fix-videos` : before generation/serving, re-encode MP4 files that are not H.264 to H.264/AAC. Requires `ffmpeg` and `ffprobe` ; the original is kept as a `.bak` file. Review the backup and converted file before deleting anything.

To view the standalone, read-only version, open `slides/index.html` . It references the media files next to it, so keep the HTML and its slide files together. The editable page should be used through the local server; its save and ZIP export actions require that server.

## Slide Order and Captions

`slides.md` determines the slide order. On the first run, media files are added in case-insensitive filename order. Subsequent runs preserve listed order and captions, remove entries for files no longer found, and append newly discovered files. Reorder entries in the Markdown file directly or use the editable page's drag-and-drop index.

Use Markdown image lines for slides and headings for section labels:

```
# First section
![Opening view](photo.jpg)
![A short clip](clip.mp4)

## Details
![Second view](diagram.svg)
![Notes](notes.md)
![Handout](handout.pdf)
```

Captions are editable in the index. Separate caption lines with `<br>` if desired. Clicking **Save Text and Order** writes the edits to `slides.md` ; if the file already exists, a timestamped `slides_YYYYMMDD_HHMMSS.md` backup is made first. Other unrecognized lines in `slides.md` are retained but ignored by the slideshow.

The editable view supports keyboard left/right navigation, zoom and pan for still images, and a ZIP export containing a standalone `index.html` plus the referenced media. The read-only view is suitable for browsing or copying with its media files; it has no save or export controls.

## Sample Media

The repository's `slides/` folder includes samples for the supported formats. Their sources are listed here; review each source's license and attribution requirements before redistributing the files or publishing a slideshow.

| File                          | Source                                    |
|-------------------------------|-------------------------------------------|
| <code>sample-png.png</code>   | Wikimedia, PNG transparency demonstration |
| <code>sample-apng.apng</code> | Wikimedia, animated beach ball            |
| <code>sample-jpg.jpg</code>   | Wikimedia, JPEG flower example            |
| <code>sample-jpeg.jpeg</code> | Google WebP gallery image                 |
| <code>sample-jfif.jfif</code> | Google WebP gallery image                 |
| <code>sample-gif.gif</code>   | Wikimedia, Newton's cradle animation      |
| <code>sample-webp.webp</code> | Google WebP gallery image                 |
| <code>sample-avif.avif</code> | link-u/avif-sample-images, fox image      |
| <code>sample-bmp.bmp</code>   | filesamples.com                           |
| <code>sample-ico.ico</code>   | python.org favicon                        |
| <code>sample-svg.svg</code>   | Wikimedia, SVG logo                       |
| <code>sample-mp4.mp4</code>   | test-videos.co.uk, Big Buck Bunny sample  |